
My Real Intuition for Next Permutation

Think of permutation like a number:

$d_0 d_1 d_2 d_3 \dots d_{n-1}$

with place values:

$10^{(n-1)}, 10^{(n-2)}, 10^{(n-3)}, \dots, 10^0$

So changing a digit on the left creates a bigger jump, and changing a digit on the right creates a smaller jump.

Suppose we have two indices:

$i < j$

If we can create a greater permutation by changing either i or j , then choosing j gives less change because j has smaller place value.

Example:

13542

Changing index 1:

13542 $\rightarrow$ 14532

Changing index 3:

13542 $\rightarrow$ 13552

The second change is smaller because it happens more to the right.

So for immediate next permutation:

Try to fix/change as far right as possible.

This suggests starting from the rightmost side.

But a greater permutation only forms when we swap a digit with a bigger digit on its right.

Suppose we want to swap index k with some right-side index j :

$k < j$

For the number to become greater:

$\text{nums}[k] < \text{nums}[j]$

If instead:

$\text{nums}[k] \geq \text{nums}[j]$

then swapping will not create a greater permutation at position k .

So while traversing from right, we are basically trying to find the first place where:

$\text{nums}[i-1] < \text{nums}[i]$

because this means:

$\text{nums}[i-1]$ can be increased using some digit on its right.

Now while moving from right to left, if we keep seeing:

$\text{nums}[i-1] \geq \text{nums}[i]$

then looking from right to left, the values are increasing:

2 -> 4 -> 5

For example:

1 3 5 4 2

5 4 2

From right to left:

2, 4, 5

This is increasing.

That means from left to right, the suffix is decreasing:

5, 4, 2

A decreasing suffix is already the maximum possible arrangement.

So we do not need to fix it yet. We keep moving left.

When this increasing pattern from right to left breaks for the first time:

2 -> 4 -> 5 -> 3

Break happens at:

3

Array:

1 3 5 4 2

^

pivot

This pivot is not magic.

It appears because:

Everything to its right is already maximum,

and this is the first position from the right
that can still be increased.

Now we have two possible regions:

left section : [0 ... pivot-1]

right section : [pivot+1 ... n-1]

We choose from the right section because we want the immediate next permutation, meaning smallest possible jump.

So we swap pivot with the just greater digit from the right section.

Example:

1 3 5 4 2

Pivot:

3

Right section:

5 4 2

Just greater than 3 is:

4

Swap:

1 4 5 3 2

Now the part I missed earlier:

After swapping, the prefix has already been minimally increased:

1 4

Now to get the immediate next permutation, the remaining right section should be as small as possible.

Current suffix:

5 3 2

Smallest form:

2 3 5

Since this suffix is decreasing, we can simply reverse it.

Final:

1 4 2 3 5

Final Recall

Start from right because right-side changes cause smaller jump.

Keep moving while $\text{nums}[i-1] \geq \text{nums}[i]$,
because that means the suffix is already maximum.

First break $\text{nums}[i-1] < \text{nums}[i]$ gives pivot.

Swap pivot with just greater digit from right section.

Reverse right section to make it minimum.